



KTU NOTES

The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPER**

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The rank of the matrix, $A =$

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & -2 \\ 2 & 4 & -3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Rank of } A = \underline{\underline{2}}$$

$$\begin{bmatrix} -1 & -4 & 9 & : & 18 \end{bmatrix} \quad R_3 \rightarrow R_3 + R_1$$

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & -4 & : & -10 \\ 0 & -3 & 10 & : & 24 \end{bmatrix} \quad R_3 \rightarrow R_3 + 3R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & -4 & : & -10 \\ 0 & 0 & -2 & : & -6 \end{bmatrix} \quad \rho(A:B) = \rho(A) = n$$

So, unique solution

$$x + y + z = 6$$

$$y - 4z = -10$$

$$-2z = -6 \Rightarrow \underline{\underline{z = 3}}$$

$$y - 12 = -10 \Rightarrow y = -10 + 12 = \underline{\underline{2}}$$

$$x + 2 + 3 = 6 \Rightarrow \underline{\underline{x = 1}}$$

$$X = \underline{\underline{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}}$$

b) Find Eigen Values And Eigen Vectors of the

$$A = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix}$$

$$A_1 = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad R_2 \leftrightarrow R_1$$

$$\sim \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2 \quad \sim \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \rho(A_1) = 2 \quad n = 3 \quad \therefore n - \rho = 3 - 2 = \underline{\underline{1}}$$

$$-x_1 + x_2 + x_3 = 0$$

$$x_2 + 2x_3 = 0$$

$$\text{Put } x_3 = k \Rightarrow x_2 + 2k = 0 \Rightarrow x_2 = \underline{\underline{-2k}}$$

$$-x_1 - 2k + k = 0 \Rightarrow -x_1 = k \Rightarrow x_1 = -k$$

$$X = \begin{bmatrix} -k \\ -2k \\ k \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix} k=1$$

To find eigen vector corresponding to $\lambda = 3$, Put

$$A_2 = \begin{bmatrix} -2 & 1 & 2 \\ -1 & -1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad R_2 \rightarrow 2R_2$$

$$\sim \begin{bmatrix} -2 & 1 & 2 \\ 0 & -3 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad R_3 \rightarrow 3R_3 + R_2 \quad \sim \begin{bmatrix} -2 & 1 & 2 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} -1 & 1 & 2 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad R_2 \rightarrow R_2 - R_1$$

$$\sim \begin{bmatrix} -1 & 1 & 2 \\ 0 & -1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \quad R_3 \rightarrow R_3 + R_2 \quad \sim \begin{bmatrix} -1 & 1 & 2 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$P(A) = 2, \quad n = 3, \quad n - r = \underline{1}$$

$$-x_1 + x_2 + 2x_3 = 0$$

$$-x_2 - x_3 = 0$$

$$\text{Put } x_2 = k$$

$$-k = x_3$$

$$-x_1 + k - 2k = 0 \Rightarrow -x_1 = k$$

$$x_1 = \underline{\underline{-k}}$$

$$x_3 = \begin{bmatrix} -k \\ k \\ -k \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} k=1 = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$$

Q. Show that the Quadratic form $4x^2 + 12x$ is Positive definite.

$$A = \begin{bmatrix} 4 & 6 \\ 6 & 13 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

(i) have no solution, if $a=b=c=1$

(ii) have many solutions if $a=b/2=c=1$

Ans:- $A = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 4 & 5 \\ 2 & 3 & 4 \end{bmatrix}$ $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ $B = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$

$$A:B = \left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 3 & 4 & 5 & b \\ 2 & 3 & 4 & c \end{array} \right] \quad \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & 1 & 2 & b-3a \\ 0 & 1 & 2 & c-2a \end{array} \right] \quad R_3 \rightarrow R_3 - R_2 \quad \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & 1 & 2 & b-3a \\ 0 & 0 & 0 & c-2a-b+3a \end{array} \right]$$

(i) $a=b=c=1$

$$(A:B) \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 1 \end{array} \right] \quad \begin{array}{l} \rho(A:B)=3 \quad \rho(A)=2 \\ \therefore \rho(A:B) \neq \rho(A) \\ \text{no solution} \end{array}$$

(ii) $a=b/2=c=1$ or $a=1, b=2, c=1$

$$(A:B) \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \rho(A)=2 < n=3$$

Consider, $(A - \lambda I)x \Rightarrow \begin{bmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{bmatrix}$

To find eigen vector corresponding to $\lambda = 8$,
eq ①.

$$A_1 = \begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 + R_1 \end{array}$$

$$A_1 = \begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix} \quad P(A) = 2, n = 3, n - r = 1$$

$$-2x_1 - 2x_2 + 2x_3 = 0$$

$$-3x_2 - 3x_3 = 0 \quad \text{Put } x_2 = \underline{k}$$

$$-3k = 3x_3 \Rightarrow x_3 = \underline{-k}$$

$$-2x_1 - 2k - 2k = 0 \Rightarrow -2x_1 = 4k \Rightarrow$$

$$x_1 = \begin{bmatrix} -2k \\ k \\ -k \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix} k = 1 = \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix}$$

$$\text{put } x_2 = k_1 \text{ \& } x_3 = k_2$$

$$4x_1 - 2k_1 + 2k_2 = 0$$

$$4x_1 = 2k_1 - 2k_2$$

$$x_1 = \frac{k_1}{2} - \frac{k_2}{2} = \frac{k_1 - k_2}{2}$$

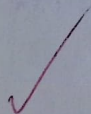
$$X_2 = \begin{bmatrix} \frac{k_1 - k_2}{2} \\ k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} \neq \begin{bmatrix} -1/2 \\ 0 \\ 1 \end{bmatrix}$$

$$X = \begin{bmatrix} -2 & 1/2 & -1/2 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$X^{-1}AX = D$$

$$\begin{bmatrix} -2 & 1/2 & -1/2 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} -2 & 1/2 & -1/2 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \underline{\underline{D}}$$



$$\begin{aligned}\frac{\partial z}{\partial x} &= -4 \cos(1^2 - (4 \times 3)) \\ (3, 1) &= -4 \cos(11)\end{aligned}$$

Find $\frac{dz}{dt}$ by chain rule, where z

$$x = t^4, \quad y = t^3$$

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial x} \times \frac{dx}{dt} + \frac{\partial z}{\partial y} \times \frac{dy}{dt} \\ &= 6xy^2 \times 4t^3 + 6x^2y \times 3t^2 \\ &= 6 \times t^4 \times t^6 \times 4t^3 + 6 \times t^8 \times 3t^2 \\ &= 24t^{13} + 18t^{13} \\ &= \underline{\underline{42t^{13}}}\end{aligned}$$

$$a) \quad \lambda(x, y, z) = f(x_0, y_0, z_0) + f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0)$$

$$f = \frac{4y}{x+z}$$

$$f_x = \frac{-4y}{(x+z)^2}$$

$$f_y = \frac{4}{x+z}$$

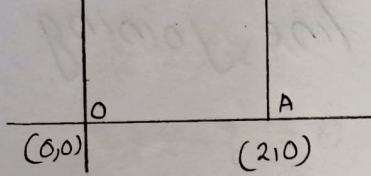
$$f_x(1,1,1) = \frac{-4 \times 1}{2^2}$$

$$= \underline{\underline{-1}}$$

$$f_y(1,1,1) = 2$$

$$f_z = \frac{-4y}{(x+z)^2}$$

$$f_z(1,1,1) = -1$$



The critical points of $f(x, y)$ in
of R are obtained by solving $f_x =$

$$f_x = 2x - 2 = 0$$

$$\Rightarrow 2x = 2$$

$$x = 1$$

$$f_y = -6y + 6 = 0$$

$$\Rightarrow 6y = 6$$

$$y = 1$$

$\therefore$ Critical point is $(1, 1)$

Consider the boundary of the region
combination of 4 line segments.

The critical point on the line joining $(0,0)$ $(2,0)$ is $(1,0)$

$\therefore$ absolute extrema can occur at boundary points $(0,0)$ $(2,0)$ and critical point $(1,0)$

line joining $(2,0)$ and $(2,2)$

$$x = 2$$

$$f(x,y) = f(2,y)$$

$$= 4 - 3y^2 - 4 + 6y$$

$$= -3y^2 + 6y$$

$$f' = -6y + 6 = 0$$

$$\Rightarrow y = 1$$

$\therefore$ The critical point on this line is

$$f(x, y) = x^2 - 2x - 2y$$

$$= x^2 - 2x$$

$$f' = 2x - 2 = 0$$

$$\Rightarrow x = 1$$

$\therefore$ The critical point on this line is
(1, 2)

$\therefore$ abs. extrema can occur on the
points (2, 2) (0, 2) and the critical,
line joining (0, 2) and (0, 0)

$$x = 0$$

$$f(0, y) = -3y^2 + 6y$$

$$f' = -6y + 6 = 0$$

$$\Rightarrow y = 1$$

$\therefore$ critical point is (0, 1)

abs. maximum value is 3.

12
7.

a) If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ find the

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}$$

b) Locate all relative extrema of
 $= 2xy - x^3 - y^2$

a) Let $a = \frac{x}{y}$, $b = \frac{y}{z}$, $c = \frac{z}{x}$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial a} \times \frac{\partial a}{\partial x} + \frac{\partial u}{\partial c} \times \frac{\partial c}{\partial x}$$

$$= \frac{\partial u}{\partial a} \times \frac{1}{y} + \frac{\partial u}{\partial c} \times \frac{-z}{x^2}$$

$$= \frac{\partial u}{\partial a} \times \frac{1}{y} - \frac{\partial u}{\partial c} \times \frac{z}{x^2}$$

$$\frac{\partial z}{\partial x} = \frac{\partial u}{\partial b} \times \frac{\partial b}{\partial x} + \frac{\partial u}{\partial c} \times \frac{\partial c}{\partial x}$$

$$= \frac{-y}{z^2} \frac{\partial u}{\partial b} + \frac{\partial u}{\partial c} \times \frac{1}{x}$$

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} + z \cdot \frac{\partial u}{\partial z}$$

$$= 0$$

$$\text{ie, } \left[x \frac{\partial u}{\partial a} \times \frac{1}{y} - \frac{\partial u}{\partial c} \times \frac{z}{x} \right] + \left[\frac{-\partial u}{\partial b} \times \frac{z}{x} + \frac{\partial u}{\partial c} \times \frac{z}{x} \right]$$

$$+ \left[-\frac{y}{z} \frac{\partial u}{\partial b} + \frac{\partial u}{\partial c} \frac{z}{x} \right]$$

$$= \underline{\underline{0}}$$

b) Critical points

$$f_x = 2y - 3x^2 = 0 \quad \text{ie, } 2y - 3x^2 = 0$$

$$\Rightarrow x = 0 \quad \text{or} \quad 2 - 3x = 0$$

$$2 - 3x = 0$$

$$3x = 2$$

$$x = \frac{2}{3}$$

$\therefore$ Critical points are $(0,0)$ $(\frac{2}{3}, \frac{2}{3})$

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

$$f_{xx} = -6x$$

$$f_{yy} = -2$$

$$f_{xy} = 2$$

$\therefore$ at $(0,0)$

$$D = (-6 \times 0) \times (-2) - 2^2$$

$$= -4$$

$D < 0 \therefore (0,0)$ is a saddle point

$$6x\left(\frac{2}{3}\right)x - 2 - 2^2$$

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$$x < 0$$

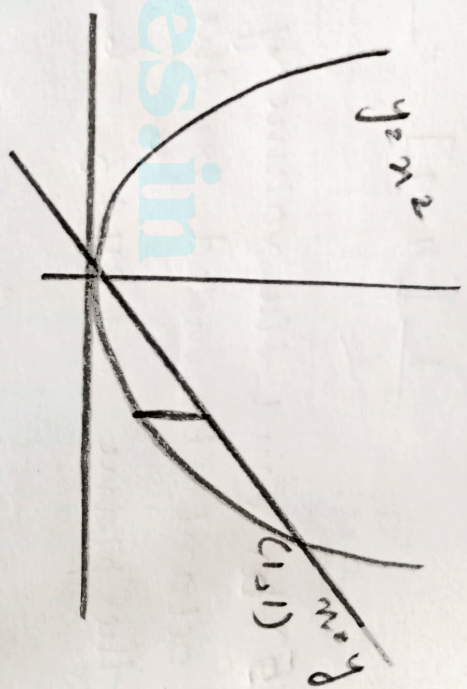
relative maximum at $(\frac{2}{3}, \frac{2}{3})$

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x^2

$$\int (x, y) dx$$

$$y = x$$



$$\begin{cases} x = x^2 \\ x - x^2 = 0 \\ x(1-x) > 0 \\ x > 0 & x < 1 \\ y > 0 & y < 1 \end{cases}$$

$$-dy dx$$

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$$= \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_0^1 = \left[\frac{1}{4} - \frac{1}{5} \right] - 0$$

$$= \frac{1}{20} //$$

6.) Evaluate $\iint_R y^2 ndA$ over the region $\{x \leq 2, 0 \leq y \leq 1\}$.

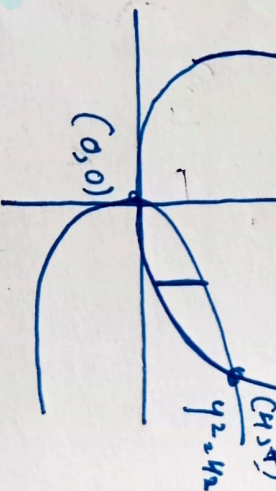
ans:

$$\iint_R y^2 ndA = \int_{-3}^2 \int_0^1 y^2 ndy dx$$

$$= \int_{-3}^2 \left[\frac{y^3}{3} \right]_0^1 ndx = \frac{1}{3} \int_{-3}^2 ndx =$$

$$= \frac{1}{6} [4 - 9] = \frac{1}{6} x - 5 = \frac{-5}{6} //$$

$$x = \int [y]^{2x} dx$$



$$x^2 = 4y$$

$$x^2 = 4 \cdot 2\sqrt{x}$$

$$x^2 = 8\sqrt{x}$$

$$x^{3/2} = 8$$

$$x^{1/2} = 2$$

$$x = 4 \Rightarrow y = 4$$

$$y : \frac{x^2}{4} \rightarrow \sqrt{4x}$$

$$x : 0 \rightarrow 4$$

$$= \int_0^4 \left[2x - \frac{x^2}{2} \right] dx$$

$$= \frac{x^2}{2} - \frac{x^3}{6} \Big|_0^4 = 4^2 - \frac{4^3}{6}$$

$$= 16 - \frac{32}{3}$$

$$= \frac{16}{3}$$

ed by the coordinate planes and

$$2y + z = 6$$

$$z = 6$$

$$z = 6 - x - 2y$$

$$\therefore z: 0 \rightarrow 6 - x - 2y$$

$$\frac{x}{0} \quad \frac{y}{\frac{5}{2}} \quad \frac{z}{3} \quad y = 6 - \frac{x}{2}$$

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$$\begin{aligned}
 & \int_0^6 \int_0^{\frac{6-n}{2}} [z]_0^{6-n-2y} dy dn = \int_0^6 \int_0^{\frac{6-n}{2}} 6-n-2y dy dn \\
 &= \int_0^6 \left[6y - ny \frac{2y}{2} \right]_0^{\frac{6-n}{2}} dn \\
 &= \int_0^6 \frac{1}{2} \left[36 - 6n - 6n + n^2 - \frac{36}{2} + 6n \right] dn \\
 &= \frac{1}{2} \int_0^6 \left[18 - 6n + \frac{n^2}{2} \right] dn = \frac{1}{2} \left[18n - 3n^2 + \frac{n^3}{6} \right]_0^6 \\
 &= \frac{1}{2} \left[18 \times 6 - 3 \times 6^2 + \frac{6^3}{6} \right] = \frac{1}{2} [36]
 \end{aligned}$$

16) a) change the order of integration and evaluate $\int_0^4 \int_{x^2/4}^{2\sqrt{x}} dy dx$.

ans: $y: \frac{x^2}{4} \rightarrow 2\sqrt{x}$

$y = \frac{x^2}{4} \Rightarrow x^2 = 4y$

$$= \int_0^4 \int_{y^2/4}^{2\sqrt{y}} dx dy$$

$$\int_0^4 [x^2]_{y^2/4}^{2\sqrt{y}} dy = \int_0^4 [2\sqrt{y} - y^2/4] dy = \frac{1}{4}$$

$$= \frac{1}{4} \left[8 \times \frac{y^{3/2}}{3/2} - \frac{y^3}{3} \right]_0^4 = \frac{1}{4} \left[\frac{16y^{3/2}}{3} \right]_0^4$$

$$= \frac{1}{12} [16 \times 2^3 - y^3]_0^4 = \frac{1}{12} [16 \times 8$$

$$= \frac{1}{12} \times 64 = \frac{16}{3} //$$

b) Evaluate $\iiint_C z dv$ where C is the
the 1st octant cut off from the cyl-
solid $y^2 + z^2 \leq 1$ and the planes $y = x$

ans: $y = x$ $y^2 + z^2 \leq 1$ $x = 0$

put $z = 0$

$y^2 \leq 1$

$y \leq 1$

$z^2 \leq 1 - y^2$

$z \leq \sqrt{1 - y^2}$

$z: 0 \rightarrow \sqrt{1 - y^2}$

ans: ~~get x elep⁴² + z² ≤ 1~~

$$= \int_0^1 \int_x^1 \left[\frac{z^2}{2} \right]_0^{\sqrt{1-y^2}} dy dx$$

$$= \frac{1}{2} \int_0^1 \int_x^1 [1-y^2] dy dx = \frac{1}{2} \int_0^1 [y$$

$$= \frac{1}{2} \int_0^1 \left[1 - \frac{1}{3} \right] - \left[x - \frac{x^3}{3} \right] dx$$

$$= \frac{1}{2} \int_0^1 \left[\frac{2}{3} - x + \frac{x^3}{3} \right] dx$$

$$= \frac{1}{2} \left[\frac{2}{3} x - \frac{x^2}{2} + \frac{x^4}{12} \right]_0^1$$

$$= \frac{1}{2} \left[\frac{2}{3} - \frac{1}{2} + \frac{1}{12} \right]$$

$$= \frac{1}{2} \left[\frac{8-6+1}{12} \right] = \frac{3}{24}$$

$$= \frac{1}{8}$$

$= \infty$
 $\therefore$ series diverges by Root test.

8. Does the series $\sum_{k=1}^{\infty} \left(-\frac{3}{4}\right)^k$ converge

Sum.

Soln. It is a geometric series with

$$r = -\frac{3}{4}$$

$$|r| = \frac{3}{4} < 1$$

$\therefore$ series converges

$$\text{Sum} = \frac{a}{1-r} = \frac{-3/4}{1 - (-3/4)} =$$

$$= \underline{\underline{-3/7}}$$

$$k \rightarrow \infty \quad a_k$$

$$= \lim_{k \rightarrow \infty} \frac{(2k-1)! (2k+1)}{(2k-1)! 2k}$$

$$= 0 < 1$$

$\therefore$ By Ratio test, the series converges

17 b)

Find the sum of the series $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$

Soln:

$$a_k = \frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{(k+1)}$$

$$S_n = a_0 + a_1 + \dots + a_n = \left(1 - \frac{1}{2}\right) + \dots + \left(\frac{1}{n} - \frac{1}{(n+1)}\right)$$

$$= 1 - \frac{1}{n+1}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1}\right) = 1$$

$$\therefore \underline{\underline{\text{Sum} = 1}}$$

$$\rho = \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{(k+1)}{3! k!}$$

$$= \lim_{k \rightarrow \infty} \frac{(k+1)}{3! k}$$

$$= \frac{1}{3} < 1$$

$\therefore$ series converges by Ratio test.

$$2) \quad a_k = \left(\frac{4k-5}{2k+1} \right)^k$$

$$\rho = \lim_{k \rightarrow \infty} a_k^{1/k} = \lim_{k \rightarrow \infty} \left(\frac{4k-5}{2k+1} \right)$$

$$= \lim_{k \rightarrow \infty} \frac{4k-5}{2k+1} = \lim_{k \rightarrow \infty} \frac{4}{2}$$

$\therefore$ series diverges by Root test.

$$\lim_{k \rightarrow \infty} a_k = \lim_{k \rightarrow \infty} \frac{k^3}{k^3+1}$$

$$a_{k+1} = \frac{(k+1)^3}{(k+1)^3+1}$$

$$a_k > a_{k+1}$$

$\therefore$ By alternating series test, $\sum_{k=1}^{\infty} a_k$

Consider the series of absolute values

$$\text{i.e. } \sum_{k=1}^{\infty} \frac{k^2}{k^3+1} \quad \therefore a_k = \frac{k^2}{k^3+1}$$

$$\text{By informal principle } \sum_{k=1}^{\infty} b_k = \sum_{k=1}^{\infty} \frac{k^2}{k^3+1}$$

harmonic series

$$\begin{aligned} \rho &= \lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \lim_{k \rightarrow \infty} \frac{k^2}{\frac{k^3+1}{1/k}} \\ &= \lim_{k \rightarrow \infty} \frac{1}{1 + 1/k^3} = 1 \end{aligned}$$

$$\therefore \sum_{k=1}^{\infty} (-1)^{k+1} \frac{k^2}{k+1} \quad \text{conditionally}$$

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$$(1+x)^{1/3} = 1 + \frac{1}{3}x + \frac{\frac{1}{3}(\frac{1}{3}-1)}{2!}x^2 + \dots$$

$$= 1 + \frac{1}{3}x + \frac{\frac{1}{3} \times \frac{-2}{3}}{2}x^2 + \dots$$

$$= 1 + \frac{1}{3}x - \frac{1}{9}x^2 + \dots$$

10. Find the Maclaurin's series of $\log(1+x)$ upto third degree term.

$$f(x) = f(0) + \frac{x}{1!}f'(0) + \frac{x^2}{2!}f''(0) + \dots$$

$$f(x) = \log(1+x)$$

$$f'(x) = \frac{1}{1+x}$$

$$f''(x) = \frac{-1}{(1+x)^2}$$

$$f'''(x) = \frac{2}{(1+x)^3}$$

$$\therefore f(x) = 0 + x \times 1 + \frac{x^2}{2!} \times (-1) + \frac{x^3}{3!} \times 2 + \dots$$

$$= x + \frac{x^2}{2} - \frac{x^3}{3} + \dots$$

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$$= \frac{1}{2\pi} \left[\frac{e^{-x}}{-1} \right]_0^{2\pi} = \frac{1}{2\pi} [e^{-2\pi} - e^0]$$

$$= \frac{1}{2\pi} [e^{-2\pi} - 1] = \frac{1}{2\pi} [1 - e^{-2\pi}]$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} e^{-x} \cdot \cos(nx) dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-x}}{1+n^2} (1 \cos nx + n \sin nx) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left\{ \frac{e^{-2\pi}}{1+n^2} [\cos n2\pi + 2\pi \sin n2\pi] \right\}$$

$$= \frac{1}{\pi} \left\{ \frac{e^{-2\pi}}{1+n^2} [\cos 2n\pi] - \frac{1}{1+n^2} [1] \right\}$$

$$= \frac{1}{\pi(1+n^2)} [1 - e^{-2\pi}]$$

$$\therefore f(x) = \frac{1 - e^{-2\pi}}{2\pi} + \sum_{n=1}^{\infty} \frac{1}{\pi(n^2+1)} (1 - e^{-2\pi}) \cos nx + \sum_{n=1}^{\infty} \frac{n}{\pi(n^2+1)} (1 - e^{-2\pi}) \sin nx$$

b) Find the half range cosine series for

$$0 \leq x \leq 1$$

Half range cosine series of $f(x)$ is,

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{l} \int_0^l f(x) dx$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$a_0 = \frac{1}{1} \int_0^1 (x-1)^2 dx$$

$$= \left[\frac{(x-1)^3}{3} \right]_0^1 = \frac{0}{3} - \frac{(-1)^3}{3} = \underline{\underline{\frac{1}{3}}}$$

$$= \frac{2x^2}{n^2\pi^2}$$

$$= \frac{4}{n^2\pi^2}$$

$$f(x) = \frac{2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2\pi^2} \cos(n\pi x)$$

20 a) Find the fourier series of the function
in $-1 \leq x \leq 1$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{2l} \int_{-l}^l f(x) dx$$

$$= \frac{1}{2} \int_{-1}^1 |x| dx = \frac{1}{2} \times 2 \int_0^1 x dx$$

$$= \left[x \times \frac{x^2}{2} \right]_0^1 = \underline{\underline{1/2}}$$

$$= \frac{2}{n^2 \pi^2} [(-1)^n - 1]$$

$$b_n = \frac{1}{\pi} \int_{-1}^1 |x| \sin(n\pi x) dx$$

$$= 0$$

$$\therefore f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2 \pi^2} ((-1)^n - 1) \cos(n\pi x)$$

20 b) Find the fourier sine series of $f(x) =$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \cos x \sin(nx) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \left[\frac{\sin(n+1)x + \sin(n-1)x}{2} \right] dx$$

$$= \frac{1}{\pi} \left\{ \pi x \frac{-1}{n+1} \cos(n+1)\pi + 0 + \pi x \frac{-1}{n-1} \right.$$

$$= \frac{1}{\pi} \times \pi \left[\frac{-1}{n+1} (-1)^{n+1} - \frac{1}{n-1} (-1)^{n-1} \right]$$

$$= (-1)^{n-1} \left[\frac{-1}{n+1} - \frac{1}{n-1} \right] = (-1)^n \left[\frac{1}{n+1} \right]$$

$$= \frac{2n}{n^2-1} (-1)^n, n \neq 1$$

When $n=1$, $b_1 = \frac{2}{\pi} \int_0^{\pi} x \sin x \cos x dx$

$$= \frac{2}{\pi} \int_0^{\pi} \frac{x}{2} \sin 2x dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x \sin 2x dx$$

$$= \frac{1}{\pi} \left[x x^{-\frac{\cos 2x}{2}} - 1 \times \frac{1}{2} x \frac{\sin 2x}{2} \right]$$

$$= \frac{1}{\pi} \left[\frac{\pi}{2} \times \cos 2\pi \right] - 0$$

$$= \frac{1}{\pi} \times \frac{\pi}{2} \times 1$$

$$= \frac{1}{2}$$

$$\therefore f(x) = \frac{-1}{2} \sin x + \sum_{n=2}^{\infty} \frac{2n}{n^2-1} (-1)^n \sin nx$$